



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering
Course Code & Title: CB3491 Cryptography and Cyber Security
Name of the Faculty member: Mrs. P.Sabeena Burvin, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / Data Encryption Standard
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 6
- **Activity Chosen:** Flipped Classroom
- **Date Implementation:** 18-08-2025
- **Justification:**

The **flipped classroom model** allows students to learn theoretical concepts independently before class and then apply them actively during in-class sessions. In this session, the focus was on understanding the **Data Encryption Standard (DES)** - one of the earliest and most influential symmetric key encryption algorithms in cryptography. Students were expected to review videos and readings on DES structure, process, and security before class. During the class session, they engaged in collaborative activities to apply the algorithm and analyze its strengths and weaknesses.

- **Time Allotted for the Activity:** 20minutes
- **Details of the Implementation:**

Pre-Class Learning

Resources Provided:

- Video Lecture: “How DES Works – Symmetric Key Encryption Explained”
- Reading Material: *Chapter on Block Ciphers (from William Stallings’ Cryptography and Network Security)*
- Supplementary Resource: Online simulation tool for DES encryption and decryption

Key Concepts Learned Before Class:

Definition of DES:

DES (Data Encryption Standard) is a symmetric-key block cipher developed by IBM in the 1970s and adopted by NIST as a federal standard for data encryption.

Key Features:

- Block size: 64 bits
- Key length: 56 bits (effective key; 64 bits including parity)
- Type: Symmetric encryption (same key for encryption and decryption)

DES Structure:

- Based on the Feistel network structure with 16 rounds of processing.
- Each round performs substitution and permutation using expansion, XOR, S-boxes, and permutation functions.

Encryption Process Overview:

- Initial Permutation (IP) → 16 Feistel Rounds → Final Permutation (FP)
- Uses Key Generation steps to derive 16 round keys from the main key.

Limitations:

- Vulnerable to brute-force attacks due to short key size (56 bits).
- Superseded by AES (Advanced Encryption Standard).

In-Class Activities

Activity 1: Step-by-Step DES Encryption Simulation

- Students worked in pairs using an online DES simulator to encrypt a sample plaintext (e.g., “HELLO123”) with a given 56-bit key.
- They observed the expansion, substitution, and permutation processes within each round.

Activity 2: Group Discussion – Security Analysis

- Groups analyzed why DES became insecure over time.
- Topics discussed: key length vulnerability, exhaustive key search, differential and linear cryptanalysis.

Activity 3: Comparison Task

- Each group compared DES with AES and 3DES in terms of speed, security, and key management.
- Findings were presented briefly to the class.

Learning Outcomes

After the flipped classroom session, students were able to:

- Explain the architecture and working principles of DES.
- Demonstrate encryption and decryption using DES structure.
- Analyze the security limitations of DES and understand why AES replaced it.
- Apply conceptual knowledge to simulate real encryption operations.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO12
CO1	3	2	1	2	-	-	-	1	1	1

(1 – Low

2 – Moderate

3 – High

PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO8	PO9	PO12
	3	2	1	2	1	1	1
	Basic Computer Science Engineering and mathematics Knowledge for performing Symmetric encryption is highly correlated	Analyzing various Symmetric encryption algorithms is highly correlated	Developing Symmetric encryption systems with basic encryption mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated	The students will adhere to the ethics.	The students will work effectively as an individual and team with how to encrypt and decrypt a message using Symmetric Encryption Algorithms.	Updating the Symmetric encryption algorithms to the current trends are moderately correlated

- Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ Feedback of practice from students and other stakeholders:
 - Student felt good, since they can study at their own pace/time.
 - They felt that through such learning, they can explore more.
- ❖ Benefit of the practice:
 - More one-to-one time between teacher and student.
 - More collaboration time for students.
 - Students learn at their own pace.
 - Practical things – like missing class due to illness – become less problematic.
 - It encourages students to come to class prepared.
- ❖ Challenges faced in implementation:
 - Relies on student preparation – few students did not come prepared.
 - The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a normal class.

References:

- ❖ <https://www.nyu.edu/faculty/teaching-and-learning-resources/strategies-for-teaching-with-tech/flipped-classes/implementing-a-flipped-class.html>
- ❖ <https://bokcenter.harvard.edu/flipped-classrooms>